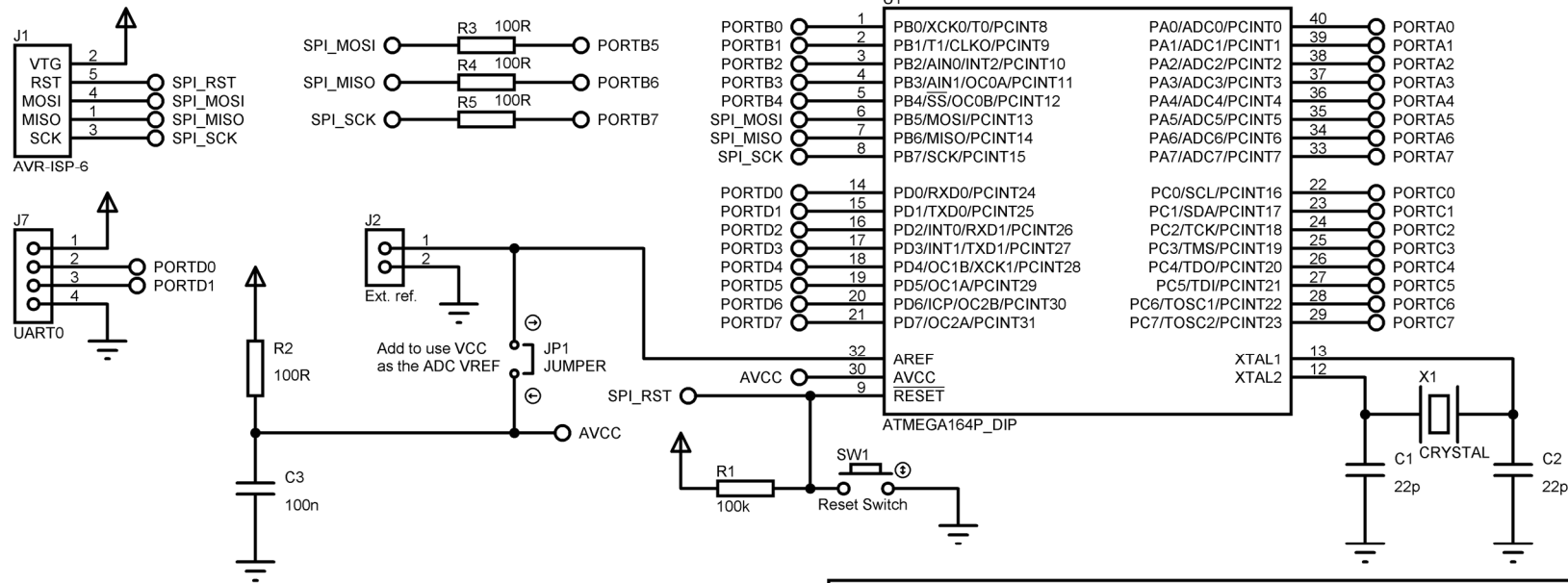


AVR Project PCB, Richard Reeves, 2018



Circuit Notes

AVR:
This PCB will accept the following: ATmega164P, ATmega324P, ATmega644P and ATmega1284P

Power Supply:
U2 is an adjustable voltage regulator. With JP2 fitted, R7 is shorted and the PCB will run at 3.3V. To run the PCB at 5V, don't fit JP2. Unregulated DC enters through J8. D2 offers some reverse-voltage protection, but care should be taken anyway when connecting power. The LM317T typically requires its input voltage to be 2V higher than its output voltage (called dropout voltage). This does not include the voltage lost across D2.

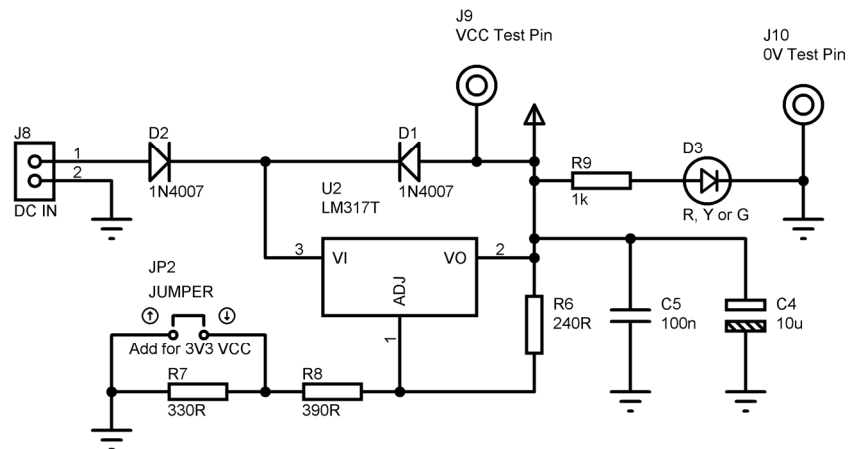
Programming:
J1 connects to the AVR programmer, with R3 - R5 providing some current limiting should the shared pins on PORTB have other peripherals connected. Programming will not work if J1 is soldered the wrong way around.

Clock:
The crystal, X1, is the clock. The frequency is by choice with a maximum of 20MHz as stated in the AVR's datasheet. Note that C1 and C2 are pF.

ADC Reference:
With JP1 fitted, AVCC (filtered by R2 and C3) is the ADC reference (AREF). This is how the lab PCBs are designed. To use an external reference (such as a precise bandgap diode), remove JP1 and connect the external reference to J2.

I/O Ports:
All 32 I/O pins are brought out through 10-way headers. J3 is PORTA, J4 is PORTB, J5 is PORTC and J6 is PORTD. 0V and VCC are available on each header. A separate file with a template for students to design PCBs to fit on top this PCB is also provided.

UART:
UART0 is also brought out for serial comms and debugging through J7. The leads are not the same connections as the lab PCBs (intentional). The wiring diagram is provided elsewhere.



UART0 is brought out on a four-pin header (a Molex KK connector). The FTDI lead to be used with these PCBs are not the same as those used in the EE2EDP AVR labs. These have four wires – the fourth wire is VCC so the circuitry in the FTDI lead runs at the same supply voltage as your circuit.

The header should be soldered to the PCB with the plastic back towards J6.

The FTDI lead (TTL-232RG-VIP-WE) has six wires, four of which you will need. The other two (the brown and green ones) are for flow control and can be trimmed.

The other four wires are the following:

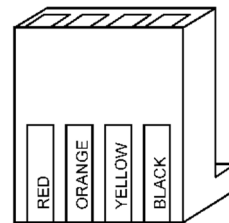
Black: 0V

Orange: Serial data from lead to AVR (i.e. **O**range for **O**utput)

Yellow: Serial data from AVR to lead

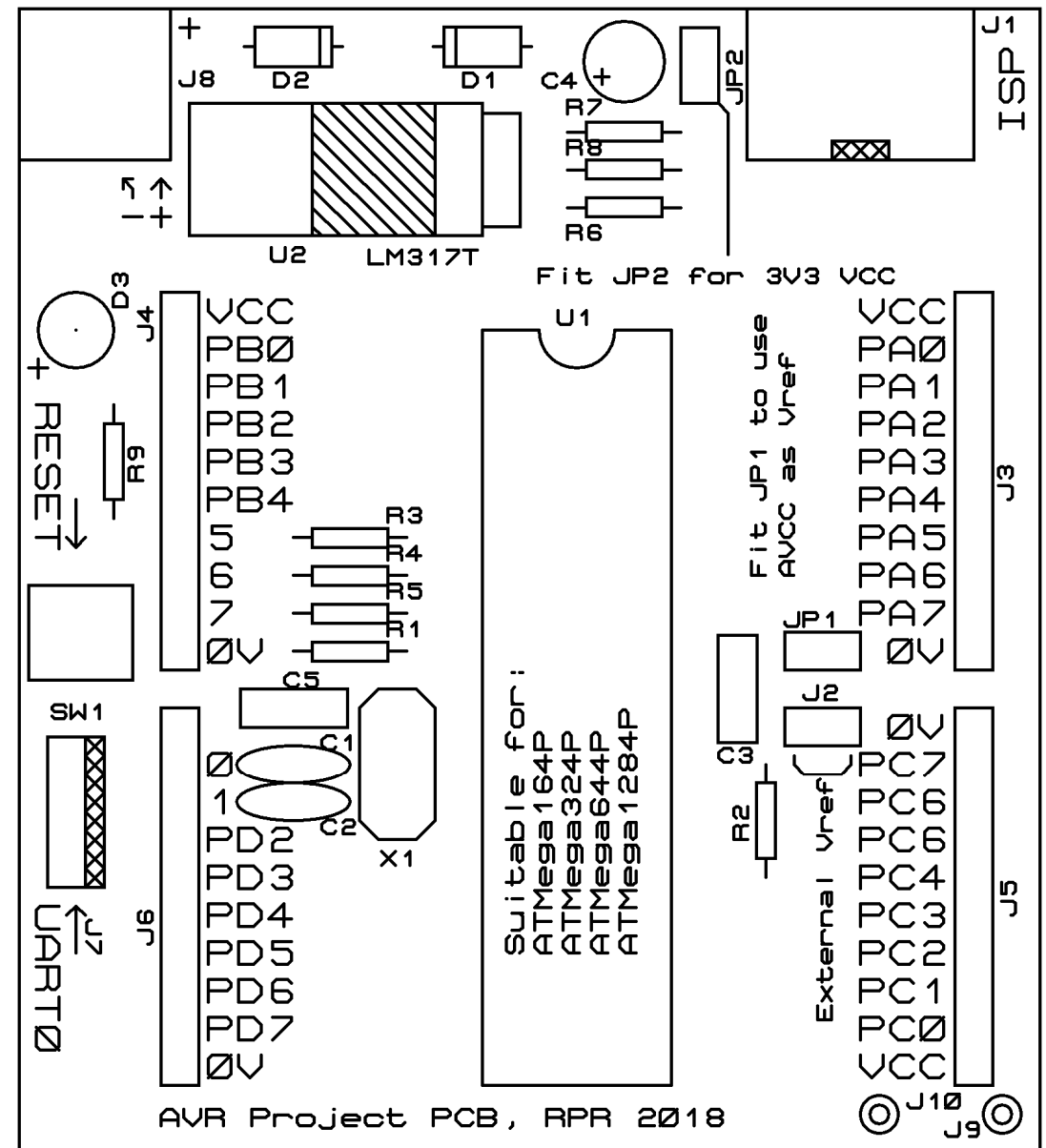
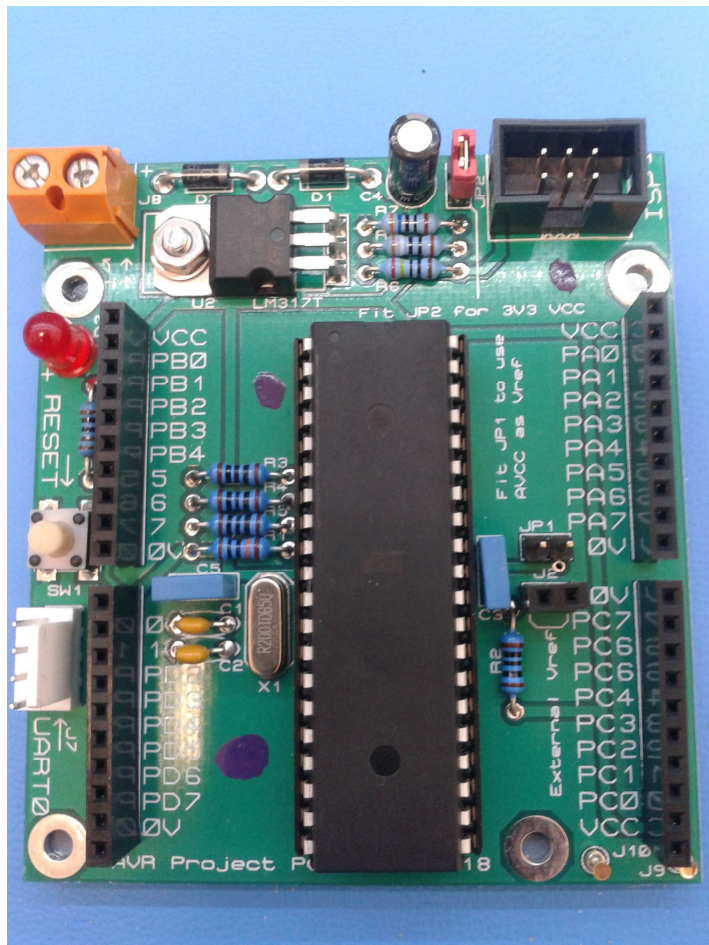
Red: VCC from your PCB

These wires will need Molex KK crimps attached. This can be done by either soldering or by using a crimping tool. Both the crimps and the crimping tool live in Richard Reeves's office. The housings for the crimp terminals have a small latch at the bottom that locks it in place when mated with the header. The crimped wires push in from above and will click into place. They can be fiddly sometimes, though. Below is a picture showing the correct colours.



The PCB's silkscreen is shown opposite. You have been taught how to solder properly so the completed board will be tidy, just like the one below.

You should know what order to solder the components and why.



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Soldering Order	Reference	Value	Category	Notes
1	R1	100k	Resistor	
2	R2	100R	Resistor	
3	R3	100R	Resistor	
4	R4	100R	Resistor	
5	R5	100R	Resistor	
6	R6	240R	Resistor	
7	R7	330R	Resistor	
8	R8	390R	Resistor	
9	R9	1k	Resistor	
10	D1	1N4007	Diode	Note the Polarity
11	D2	1N4007	Diode	Note the Polarity
12	X1	User Choice	Crystal	
13	U2	LM317T	Integrated Circuit	This lies flat and is secured with an M3x6mm bolt
14	J9	VCC Test Pin	Pin	
15	J10	0V Test Pin	Pin	
16	U1	40-Way IC Socket	Integrated Circuit	Solder the socket in, not the IC
17	C1	22p	Capacitor	
18	C2	22p	Capacitor	
19	C3	100n	Capacitor	
20	C5	100n	Capacitor	
21	SW1	FSM10JH	Reset Switch	
22	J2	2-Way Header	Aref Header	
23	J3	10-Way Header	Port Header	
24	J4	10-Way Header	Port Header	
25	J5	10-Way Header	Port Header	
26	J6	10-Way Header	Port Header	
27	J1	6-Way Box Header	AVR ISP Header	Make sure it is the correct way around
28	JP1	2-Way Jumper Pins	Aref Jumper	
29	JP2	2-Way Jumper Pins	VCC Jumper	
30	D3	LED (R, Y or G)	Diode	Note the Polarity
31	J8	2-Way Terminal Block	DC IN	
32	J7	4-Way Molex KK Header	UART0 Header	
33	C4	10u	Capacitor	Note the Polarity (it may explode if installed backwards)